



PRODUCT NAME

High-performance Glass Cleaner Formula

Description

A professional glass cleaner formulation designed to provide a streak-free finish with fast-drying properties, ensuring crystal-clear surfaces.

Technical Specifications

pH Level:	7
Viscosity:	Low
Shelf Life:	24 months
Batch Size:	100L - 1000L
Processing Time:	1 - 2 hours
Temperature:	Ambient (20-25°C)
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Mixing vessel, Stirrer, pH meter
Certification:	Complies with relevant glass and surface cleaner standards

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Deionized Water	Aqua	75.0%	Solvent
2	Isopropyl Alcohol	Isopropyl Alcohol	15.0%	Solvent, Cleaning agent
3	Ethylene Glycol Monobutyl Ether	Butoxyethanol	5.0%	Solvent, Degreasing agent
4	Ammonium Hydroxide	Ammonium Hydroxide	2.0%	pH adjuster, Cleaning agent
5	Sodium Lauryl Sulfate	Sodium Lauryl Sulfate	1.0%	Surfactant
6	Fragrance	Fragrance	0.5%	Scent
7	FD&C Blue No. 1	CI 42090	0.1%	Color
8	Preservative	Preservative	0.4%	Preservation
Total			100%	

Manufacturing Process

Phase 1: Aqueous Phase

1. Add Deionized Water to a clean mixing vessel.
2. Introduce Sodium Lauryl Sulfate and mix until completely dissolved.

Phase 2: Solvent Phase

1. Gradually add Isopropyl Alcohol and Ethylene Glycol Monobutyl Ether to the mixture.
2. Stir continuously to ensure even distribution of solvents.

Phase 3: Final Adjustment Phase

1. Add Ammonium Hydroxide to adjust the pH to approximately 7.
2. Incorporate the Fragrance and FD&C Blue No. 1 until well blended.
3. Finally, add the Preservative and mix thoroughly.

Required Equipment

Mixing vessel, Stirrer, pH meter

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Complies with relevant glass and surface cleaner standards

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100L - 1000L.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Spray onto glass surfaces and wipe with a clean, dry cloth for a streak-free finish. Avoid overuse to ensure fast drying.